A3 Q1

Thursday, October 29, 2026 8:41 AM

1. R^2 is exactly the square of the sample corr. coef., but only in SLR, R^2 considering multiple values will instead describe the combination of explanation in all variables. $\hat{\rho}$ measures the linear relationship b/w the dependent and indep variables, and R^2 (and therefore $\hat{\rho}^2$) measures the ratio b/w how well this linear relationship and dependent var. (combined, SS(Reg)) have w/ the indep. variable (SS(Tot))

$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

and

$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$

$\hat{y}_i = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 x_i$

$\hat{y}_i - \bar{y} = \hat{\beta}_1 x_i - \hat{\beta}_1 \bar{x}$

$\hat{y}_i - \bar{y} = \hat{\beta}_1 (x_i - \bar{x})$

Substitute into R^2 formula

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\begin{aligned}
R^2 &= \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 \Big/ \sum_{i=1}^{n} (y_i - \bar{y})^2 \\
&= \sum_{i=1}^{n} (\hat{\beta}_1 (x_i - \bar{x}))^2 \Big/ \sum_{i=1}^{n} (y_i - \bar{y})^2 \\
&= \hat{\beta}_1^2 \sum_{i=1}^{n} (x_i - \bar{x})^2 \Big/ \sum_{i=1}^{n} (y_i - \bar{y})^2 \\
&= \left[\frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \right]^2 \left[\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \right] \\
&= \frac{\left[\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \right]^2 \sum_{i=1}^{n} (x_i - \bar{x})^2}{\left(\sum_{i=1}^{n} (x_i - \bar{x})^2 \right)^2 \sum_{i=1}^{n} (y_i - \bar{y})^2} \\
&= \frac{\left[\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \right]^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2 \sum_{i=1}^{n} (y_i - \bar{y})^2} \\
&= \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})^2}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2 \sum_{i=1}^{n} (y_i - \bar{y})^2}} \\
&= \hat{\rho}^2
\end{aligned}
$$

$\therefore$ We have shown that $R^2 = \hat{\rho}^2$ for SLR.

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